Android physical-device validation through manual review

Checkpoint: **26 October 2026, Asia/Kolkata**. Device: Motorola Edge 50 Neo, Android 16 / API 36, arm64-v8a. This records a controlled hostel demo, not a classroom accuracy study or production approval.

Completed sequence and evidence

Step	Observed outcome
Install and launch	Verified native development APK installed through ADB; SPVS sign-in UI observed
Network	Metro initially reached through USB, then LAN; API reachable over the campus Wi-Fi connection
Wi-Fi collection	In-app setup check displayed actual AP identifiers and RSSI on the phone
Separate test account	Dedicated Android demo student enrolled in CS401; original iPhone account retained
Sign-in issue	Email contained an embedded space after a dot; server schema rejected it; correction produced the Android home screen
Device registration	25 October 20:27:43 IST; native P-256 registration, client reports android-hardware storage
Face enrollment	IMAGE_QUALITY_LOW at 20:29:31 IST, followed by successful enrollment at 20:30:24 IST
Attendance submission	25 October 20:33:45 IST; identity/liveness/GPS passed; scan available; result PENDING_REVIEW
Review request	Student opened a dispute for the attempt
Faculty correction	26 October about 03:28 IST; one manual PRESENT record with source CORRECTION
Dispute resolution	One dispute resolved; faculty audit events recorded
Student API history	Exactly one PRESENT/CORRECTION record; original attempt remains PENDING_REVIEW
Android history UI	Verified after Wi-Fi reconnection: official PRESENT/CORRECTION above original PENDING REVIEW attempt

The authenticated local faculty API was exercised by the assistant at the user's request. The correction reason explicitly identifies a controlled Android hostel workflow test with classroom Wi-Fi validation still outstanding. This was not a real institutional attendance approval, automatic acceptance, or a completed faculty-browser click-through test. The faculty dashboard was started and opened, but remained at sign-in while the API review was performed.

The database confirms one correction from UNRECORDED to PRESENT, a faculty actor, a non-empty reason, ATTENDANCE_CORRECTED and DISPUTE_RESOLVED audit events, and a resolved dispute. The original automated

decision was not rewritten.

Wi-Fi collection versus classroom discrimination

The user clarified that the test location was a **hostel room**, with its AP outside. The intended classroom reportedly has an AP inside; that classroom has not yet been tested. SSIDs, raw BSSIDs and exact coordinates are omitted here.

Same observed AP	Initial hostel-room sample	Corridor sample	Repeat room sample
AP A, 2.4 GHz	-75 dBm	-52 dBm	Not observed in this scan
AP B, 5 GHz	-89 dBm	-67 dBm	-79 dBm
Total displayed APs	2	6	1

Corridor signals were 23 dB and 22 dB stronger for these APs. The user scanned outside and returned without rescanning; the retained UI was inspected before the repeat inside check. Location labels follow the user's procedure, not independent position tracking. UI inspection times are not radio timestamps.

An offline rule check used these measured RSSI values, minima equal to the first inside readings, and identical **synthetic valid GPS** inputs to isolate Wi-Fi. Both samples returned PRESENT with match ratio 1.0. That experiment made no attendance submissions or database changes. It shows that this minimum-RSSI rule cannot separate these samples; it neither proves nor disproves performance with different AP placement in the actual classroom. One scan is not calibration.

No hostel APs or thresholds were configured as classroom references. The actual classroom needs confirmed AP identities, repeated samples across seats, doorway, corridor and adjacent room, then evaluation on fresh samples.

Actual Android attendance evidence

The submitted attempt reported Wi-Fi available with source scan, GPS distance 17.4 m, horizontal uncertainty about 28.1 m, location age 0.5 s, and liveness PASSED. Mandatory face matching and device-signature checks completed before the presence decision. Provider attestation was not configured.

The result was PENDING_REVIEW with WIFI_UNAVAILABLE because the classroom has zero configured AP references. In this case the reason means comparison was unavailable, not radio collection failure. This ambiguous reason code is retained in the historical attempt; no weaker policy was enabled to force approval.

After the labelled demo correction, the official status is PRESENT with source CORRECTION. Original attempt reasons remain visible. The history UI still uses generic pending-review explanatory text and may offer a review action for the original attempt; the official-record section is authoritative. Improving that resolved-state wording/action is a remaining UX item.

Sign-in fix and network recovery

The invalid email was confirmed from the non-password input and validated against the actual server schema. The intended credentials independently returned HTTP

200. Removing only the embedded space allowed phone login; no password was logged.

The mobile form now requests an email keyboard and email autofill hints. Sign-in rejects embedded whitespace locally and translates server email/password field validation into clear messages. Authentication errors remain distinct. Four new regressions bring the mobile suite to 46 passing tests; TypeScript passes.

The history screen initially waited for responses while the phone had Wi-Fi enabled but was not connected. API/student-history checks succeeded on the Mac. After the user reconnected Wi-Fi, the actual phone UI displayed the saved record and original attempt. This confirms recovery for this case, not a comprehensive network interruption/timeout test. Requests can still wait without a clear timeout/loading indication; that remains a UX/reliability item.

What remains

Actual classroom automatic acceptance/rejection, broader Android/iPhone device coverage, false-match/spoof evaluation, provider attestation and production operations remain unverified. Hardware storage is client-reported, not independently attested. No new claim of classroom-level accuracy follows from the demo.

Sanitized database/UI-derived evidence is in android-physical-validation.json. Raw UI snapshots, router identifiers, local credentials, templates and private keys are excluded from publication. The earlier Mac build record remains the dated native compilation record; this document supersedes its then-pending installation and initial physical-flow items.